1.2 How Thinly Spread Are the Primes?

WORKSHEET

The question

STEP III 2016, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s3-q5

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Part (iv) asks you to prove a statement for *all* $n \geq 2$. Part (iii) assumes the statement for every k up to $2m$. What kind of induction is that?

STAGE 1 A binomial coefficient is not too big

Covers: Part (i)

Ask yourself Where does 2^{2m+1} come from? Put $x = 1$ in the binomial expansion of $(1+x)^{2m+1}$.

Ask yourself All the terms of that sum are positive. Which two of them are equal to $\binom{2m+1}{m}$?

Pause and try Twice $\binom{2m+1}{m}$ is at most the whole sum, and you need a strict inequality. Is there a term you have not yet used?

STAGE 2 Which primes divide that binomial coefficient?

Covers: Part (ii)

Ask yourself Write $\binom{2m+1}{m}$ with factorials, and multiply up so that there are no fractions.

Ask yourself Take a prime p with $m+1 < p \leq 2m+1$. Does p divide $(2m+1)!$? Can it divide $m!\,(m+1)!$, whose factors are all at most $m+1$?

Pause and try So each such prime divides $\binom{2m+1}{m}$. Why does their product divide it too, and how does part (i) then finish the part?

STAGE 3 Splitting the product

Covers: Part (iii)

Ask yourself Split the primes up to $2m+1$ into those up to $m+1$ and those above $m+1$. What does that do to $P_{1,2m+1}$?

Pause and try Use the hypothesis of this part on one factor and part (ii) on the other. Write both bounds as powers of 4.

STAGE 4 All n

Covers: Part (iv)

Ask yourself Use strong induction on n. Part (iii) handles which values of n?

Pause and try For the other values of n: can an even number greater than 2 be prime? What does that tell you about $P_{1,n}$ compared with $P_{1,n-1}$?

What's really going on

From 4^n to the Prime Number Theorem

Take logarithms of the result: $\ln P_{1,n} < n \ln 4$, that is,

$$\sum_{p \le n} \ln p < (\ln 4)\, n \approx 1.39\, n.$$

In the 1850s Pafnuty Chebyshev proved bounds of exactly this kind, above *and below*, showing that $\sum_{p \le n} \ln p$ grows in proportion to n. Gauss, aged about 15, had already guessed from tables that the number of primes up to n is roughly $n / \ln n$. This became the *Prime Number Theorem*, finally proved in 1896 by Jacques Hadamard and Charles de la Vallée Poussin using complex analysis. In its logarithmic form it says

$$\sum_{p \le n} \ln p \sim n,$$

so the product of the primes up to n is roughly e^n. Our bound of 4^n is not far off at all: Figure 1.2.1 shows all three.

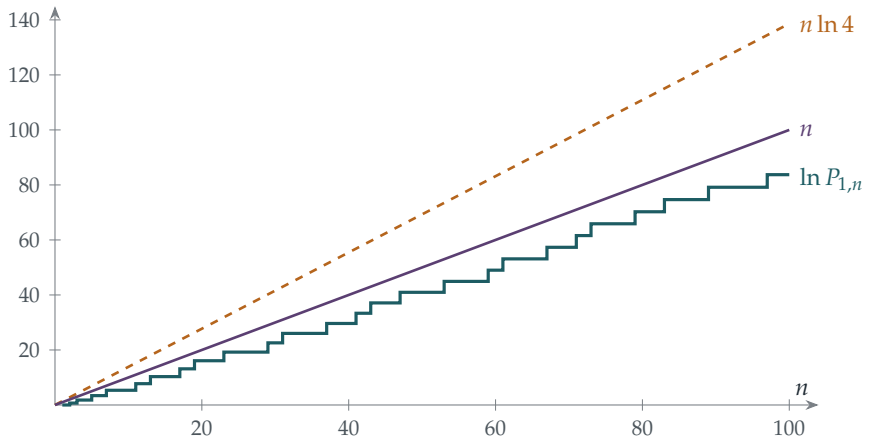


Figure 1.2.1 The logarithm of the product of all primes up to n (teal), jumping by $\ln p$ at each prime p. This section proves it stays below $n \ln 4$ (amber, dashed); the Prime Number Theorem says it grows like n (plum).

Bertrand's postulate

In 1845 Joseph Bertrand conjectured that for every $n \ge 1$ there is always a prime between n and $2n$. Chebyshev proved it in 1852. Eighty years later, Paul Erdős, aged 19, found a beautiful elementary proof. Its key ingredients are this question's lemma and a careful look at which primes divide the *middle* binomial coefficient $\binom{2n}{n}$. Martin Aigner and Günter Ziegler chose it as Chapter 2 of *Proofs from THE BOOK*, a collection of the most elegant proofs in mathematics.

Where this goes next

The precise distribution of the primes is controlled by the zeros of the Riemann zeta function,

$$\zeta(s) = 1 + \frac{1}{2^s} + \frac{1}{3^s} + \cdots,$$

a function you will meet in a much humbler guise in Section 2.4, where $\zeta(2) = \pi^2/6$. The *Riemann Hypothesis*, which predicts exactly where those zeros lie, would pin down the error in the Prime Number Theorem as tightly as possible. It is perhaps the most famous unsolved problem in mathematics, and one of the Millennium Prize Problems.

Try this next

1. **Feel the numbers.** Compute $P_{1,n}$ for $n = 10, 20, 30$ and compare it with 4^n and with e^n. (A calculator is fine here.)

2. **The middle binomial coefficient.** Show that $\binom{2m}{m}$ is the largest of the $2m+1$ terms in the expansion of $(1+1)^{2m}$, and deduce that $\binom{2m}{m} \geq \dfrac{4^m}{2m+1}$.

3. **Not too many primes.** Let $\pi(n)$ be the number of primes up to n. Every prime between $\sqrt{n}$ and n is bigger than $\sqrt{n}$, and their product is at most $P_{1,n} < 4^n$. Deduce that
$$\pi(n) - \pi(\sqrt{n}) < \frac{2\ln 4}{\ln n}\, n,$$
and hence that $\pi(n) < 2.8\, n/\ln n + \sqrt{n}$. Compare this with the Prime Number Theorem's $\pi(n) \approx n/\ln n$.

4. **Check Bertrand.** Verify Bertrand's postulate for $n = 1, 2, \ldots, 50$ by finding a prime between n and $2n$ in each case. Can you do it efficiently using a short list of primes, each less than twice the one before?

Further reading

- **Aigner & Ziegler, *Proofs from THE BOOK*,** Chapter 2, "Bertrand's postulate".
- **Marcus du Sautoy, *The Music of the Primes*** (Fourth Estate). A popular history of the Prime Number Theorem and the Riemann Hypothesis.
- **G. J. O. Jameson, *The Prime Number Theorem*** (Cambridge University Press). Rigorous but gentle, and written for undergraduates; the first chapters need only A-level calculus.

The full guided solution is in Section 1.2:
https://beyondstep.keithmcnulty.org/ch01/prime-distribution.html